Homework 5 Answers

BSTA 550

Questions

1. (a) parameters: $\sum_{i=1}^{m} n_i$ and p
   (b) Not given
   (c)
   $$p(1-p)\sum_{i=1}^{m} n_i$$

2. (a) Exponential distribution
   (b) $\lambda = 240$
   (c) Not given

3. 0.252

4. Not given

5. $f_Z(z) = nF_X(z)^{n-1} f_X(z)$

6. - (b) $P(X < Y) = 0.099995$
   - (d) $f_Z(z) = \frac{11-2z}{5} e^{-2z}$ for $0 \leq z \leq 5$

7. 2 cases: when $0 < z < 1$ and $1 \leq z \leq 2$